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Introduction to

Python Programming

Introduction to Python Programming

SENIOR CONTRIBUTING AUTHORS
UDAYAN DAS, SAINT MARY'S COLLEGE OF CALIFORNIA
AUBREY LAWSON, WILEY
CHRIS MAYFIELD, JAMES MADISON UNIVERSITY
NARGES NOROUZI, UC BERKELEY

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OpenStax
Rice University
6100 Main Street MS-375
Houston, Texas 77005

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DIGITAL VERSION ISBN-13 **978-1-961584-45-7**
ORIGINAL PUBLICATION YEAR **2024**
1 2 3 4 5 6 7 8 9 10 CJP 24

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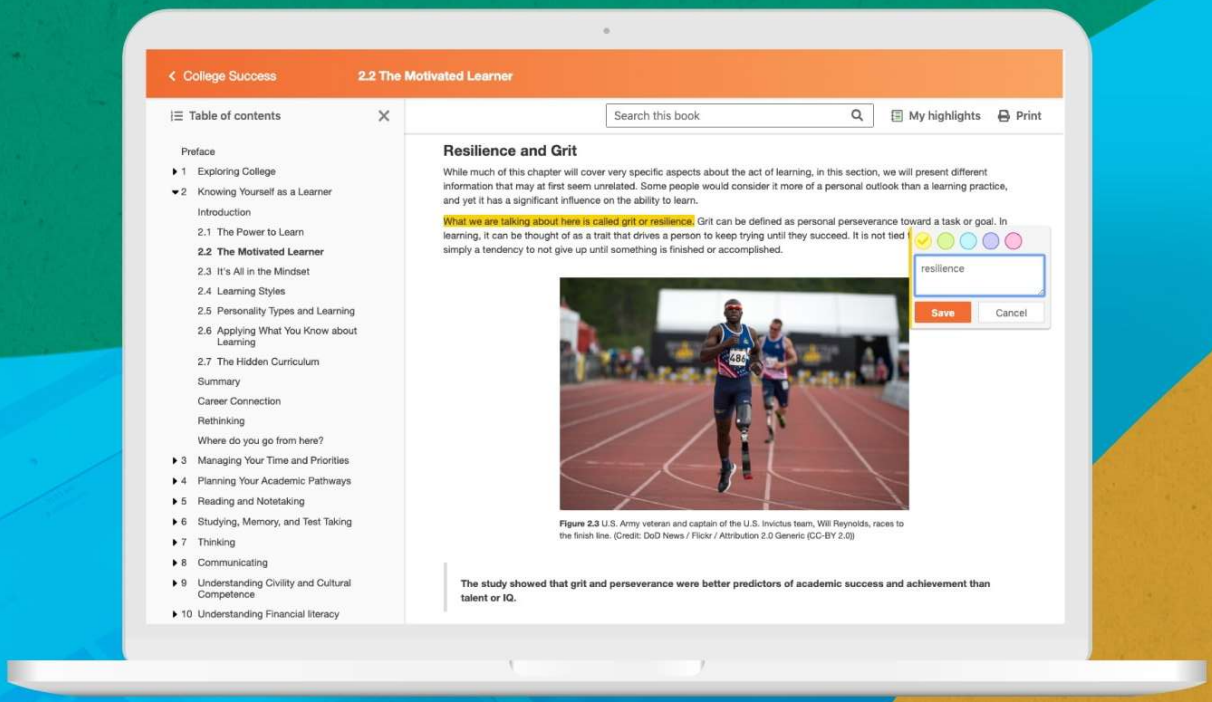

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Preface

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Format

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About *Introduction to Python Programming*

Introduction to Python Programming provides a comprehensive foundation in programming concepts and skills, and is aligned to the scope of most introductory courses. A wide array of scenarios, contexts, and problems reflect programming applications in many disciplines and careers. The offering is suitable for a diverse learner audience, including those pursuing computer science, business, science, social science, statistics, data science, and related areas of study and employment.

Introduction to Python Programming is an interactive offering that teaches basic programming concepts, problem-solving skills, and the Python language using hands-on activities. The resource includes a unique, integrated code runner, through which students can immediately apply what they learn to check their understanding. Embedded videos, critical thinking exercises, and explorations of external programming tools and activities all contribute to a meaningful and supportive learning experience.

The content is organized in chapters, with each chapter containing 6-8 sections. Each section follows the pattern:

- Learning objectives
- 1–3 subsections
- Programming practice

The learning objectives are designed to help readers identify the section's focus. Each objective completes the sentence, "By the end of this section you should be able to". The programming practice aligns with the learning objectives and gives readers an opportunity to apply concepts learned in the section.

Pedagogical Foundation

Concise text and video-based animations

Introduction to Python Programming is designed to foster active learning and student engagement. It focuses on interactivity and practice through its integrated code runner, videos, and links to external environments and activities. With that focus, the material is often more concise, with less text and more activity.

Each section's content is organized in subsections. The subsection begins with a concise introduction to the concept, with key term definitions and brief context for its relevance and importance. The concept is then explained in detail using video-based animations and interactive learning questions.

Animation videos use a step-by-step approach to show the execution of Python code. Explanations for each step describe Python syntax, show how fundamental programming concepts are used, illustrate how variables are assigned, emphasize how code executes line by line, apply problem solving to create programs, and more.

CHECKPOINT

Displaying output to the user

Access multimedia content (https://www.openstax.org/r/displaying-output)

Learning questions

After introducing a new concept and explaining the concept with a video-based animation, each subsection includes engagement in the form of learning questions. These questions reinforce the concepts taught, explain concepts in more depth, directly address misconceptions and errors commonly made by new programmers, and teach related concepts.

Explanations are provided for the incorrect answers. Incorrect answers' explanations include why the answer is incorrect and help guide the reader to the correct answer.

Incorrect answer choices typically represent a misconception or are the result of making a common mistake. Even if the correct answer is achieved, readers are encouraged to explore the explanations to gain awareness of these common misconceptions.

Programming practice exercises

Each section ends with 1 or 2 practice programs. This book includes an integrated programming environment, known as the "OpenStax Python Code Runner," which allows readers to write programs directly in the browser.